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ORIGINAL RESEARCH

Adversarial training flat-lattice transformer for named entity recognition of chinese legal texts

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Abstract
Judgment documents are the legally binding written conclusion made by the court based on the facts of the case and the law. Due to the use of professional terms and nested combinations of words, potential information of judgment documents has not been deeply excavated. Named Entity Recognition (NER) is a necessary task in Natural Language Processing (NLP), and has been widely introduced into Chinese texts processing for many years. However, the professional terms and nested words lead to the boundaries between entities being blurred, which can not accurately divide entities. In this paper, a new NER model Adversarial Training Flat-Lattice Transformer (AT-Flat) which combines adversarial training and Flat-Lattice Transformer (Flat) is proposed to weaken these problems. In AT-Flat, the Flat combines character and word information to get sequence information, and the CRF is used to output the final entity prediction results. Moreover, the key point is an adversarial training framework introduced to integrate task-shared word boundary information from Chinese Word Segmentation (CWS) task into Chinese NER task. The framework is able to filter out the noise caused by CWS task and further enhance the effect of Chinese NER task. More importantly, experiments on NER task of Chinese traffic accident and financial lending judgment documents show that our method outperforms other state-of-the-art methods. These verify that our method can effectively alleviate the problem of poor NER effect caused by professional terms and nested words. In addition, three public Chinese NER datasets were also used to evaluate our method.

Keywords Chinese judgment documents · Nested named entity recognition · Adversarial training · Flat-lattice transformer

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1 Introduction

The amount of data in the judicial field has increased rapidly in recent years. Legal professionals such as judges, lawyers, and prosecutors not only handle a large number of cases, but also need to consult numerous references. Judgment documents are the legally binding written conclusion made by the court based on the facts of the case and the law. With the development of judicial online system, nearly 100 million judgment documents have been published online in China, accumulating abundant judicial data of various cases (Tang et al. 2020). Judgment documents can help professionals get similar cases to assist in sentencing. The information mining of judgment documents can reduce the labor cost of professionals and improve legal work efficiency.

Natural Language Processing (NLP) is a common way for processing data, of which Named Entity Recognition (NER) is an indispensable research topic. NER aims to extract the words or expressions denoting specific entities from documents (Huang et al. 2020), which is composed of two parts: detecting the entity boundaries and classifying the entities into predefined categories such as Name, Judicial Organization, and Location (Tang et al. 2020). Chinese judicial NER can improve the efficiency of court operations and provide support to judges in their trial work.

Due to the differences from other languages such as English, there are significant challenges in the Chinese NER task. The reasons are as follows: first, the Chinese language doesn't have spaces between words and lacks semantic features like uppercase, which makes many NER methods are not suitable for Chinese; secondly, the Chinese dataset for NLP is small compared to the English dataset; third, there are multiple meanings of words in Chinese, and the meaning of the same word may be vary in different contexts (Chang et al. 2021).

Furthermore, judicial entities are mostly nested, making structures complicated. For example, the nested types of the legal entities in Fig. 1, include "Judicial Organization and Location", "Causing-trouble Vehicle and defendant", "Location and Defendant", and "Stolen Goods and Causing-trouble Vehicle". "成都市中级人民检察院 (Chengdu Intermediate People's Procuratorate)" is supposed to be categorized into 'Judicial Organization', but "成都市(Chengdu)" may be categorized into 'Location' in other text (Tang et al. 2020). On the one hand, it leads to a decline in the accuracy of entity recognition. On the other hand, it loses the correlation between words, which also affects the subsequent downstream tasks (Zhang et al. 2023). These entities are rare in other domains, which makes some existing NER methods ineffective for the judicial documents.

Although existing NER methods achieve great success in many domains, most of them normally ignore the nested nature of named entities (Wang et al. 2022). There are some researches on the nested NER task by applying deep learning models in resent years, such as Long Short-Term Memory (LSTM), attention mechanism, and neural layered model. Ju et al. (2018) proposed a novel neural model to identify nested entities by dynamically stacking flat NER layers. Each flat NER layer is based on a bidirectional long short-term memory (BiLSTM) layer to capture sequential context representation and a Conditional Random Field (CRF) decoder. Fei et al. (2020) introduced a dispatched attention neural model with multi-task learning for

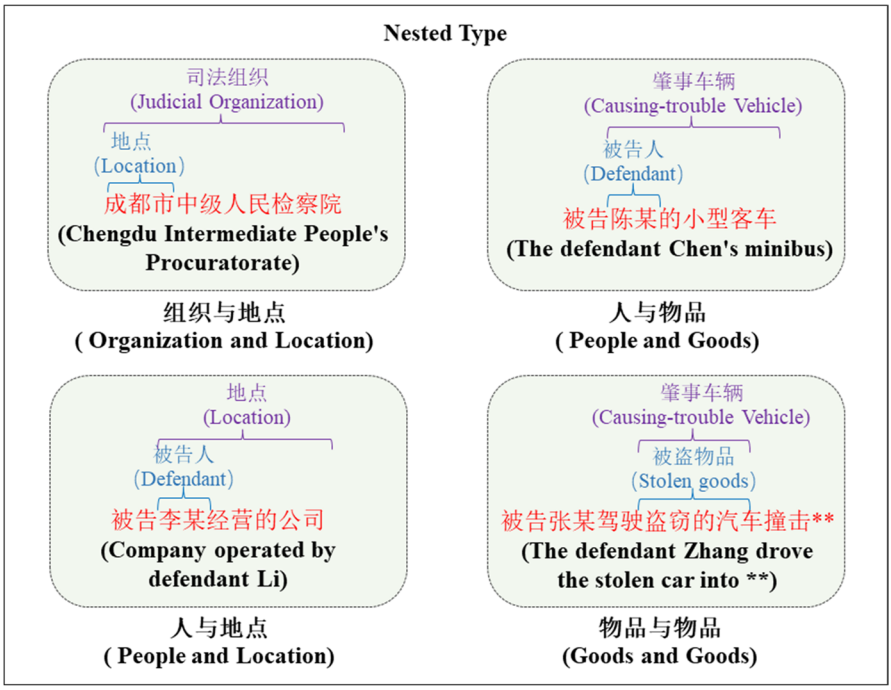

Fig. 1 Nested types of legal entities

the task, including a BiLSTM encoder, an attention network, and a CRF decoder. Wang et al. (2020) presented Pyramid, a novel layered model for nested NER. In the Pyramid, token or text region embeddings are recursively inputted into L flat NER layers, from bottom to top, stacked in a pyramid shape. They also designed an inverse pyramid to allow bidirectional interaction between layers. Although there are some methods for nested NER, few of them have been applied to the Chinese legal texts.

There are probably only few methods are effective in the Chinese judicial NER task for the above problems. As to the techniques applied in NER, there are three streams (Li et al. 2020): 1) Rule-based approaches, which do not need annotated data as they rely on hand-crafted rules; 2) Statistic-based methods, including conditional random field (Chen et al. 2006), maximum entropy model (Bender et al. 2003), and hidden Markov model (Morwal et al. 2012), etc. 3) Deep-learning based approaches. Rule-based and statistic-based methods are very corpus-dependent and the experts need to manually extract features from the dataset. Due to domain-specific rules and incomplete dictionaries, high precision and low recall are often observed from these methods, and they cannot be transferred to other domains (Li et al. 2020). In recent years, with the development of deep learning, deep-learning NER models have received more attention and achieved better results. However, deep-learning methods predict each character with a given feature independently without considering the previously predicted labels, which may lead to invalid predicted label sequences and thus reduce the accuracy of entity recognition results (Tang et al. 2020). These NER methods mentioned above are rarely applied to the judicial field.

Based on these problems, this paper proposes a new NER model Adversarial Training Flat-Lattice Transformer (AT-Flat) that combines adversarial training and Flat-Lattice Transformer (Flat). We summarize the major contributions as follows:

•To improve the performance of Chinese judgment documents NER task, a word-enhanced model combining character and word information is built. The fusion of character and word can enhance the entity's information, and provide more help for NER task.

•In order to analyze judgment documents, the adversarial framework is introduced into the legal field. The adversarial training framework extractes the common features of CWS and NER task, while the noise of CWS task can be filtered. It makes the boundary of legal entities more clear and improve the effect of NER. To the best of our knowledge, this is the first work introducing adversarial training to process the nested NER of Chinese judgment documents.

•Applying the proposed method to the NER task of real Chinese judgment documents and public Chinese NER datasets, and the results show that our method outperforms other state-of-the-art methods.

This paper is organized as follows. Section 2 reviews the related works in judgment documents analysis. Section 3 presents the proposed model architecture and detailed descriptions. The experimental results of real datasets are discussed in Section 4. Finally, the conclusion is drawn in Section 5.

2 Related work

The processing methods for Chinese legal texts are mainly applied to judgment prediction, case text classification and information extraction. Liu and Hsieh (2006) built a domain dependent word list with the help of Chien's PAT tree-based method and HowNet, and used the resulting word list for defining relevant phrases for classifying Chinese judicial documents. Xiao et al. (2018) implemented several conventional text classification baselines for judgment prediction and experimental results have showed that it is still a challenge for current models to predict the judgment results of legal cases, especially on prison terms. Chen et al. (2020) proposed a Legal Triplet Extraction System for drug-related criminal judgment documents. The system extracts the entities and the semantic relations jointly and benefits from the proposed legal lexicon feature and multi-task learning framework. Wang et al. (2020) divided entity type into four categories, and presented a model of NER based on attention mechanism; the structure improvement of input vector fusion CNN embed mode.

In the judicial field, only a few researchs on named entity recognition for Chinese legal texts. Shi et al. (2022) established the corpus dataset of specific cause of action and proposed RoBERTa (a variant of the pre-trained language model BERT)-BiLSTM-CRF legal case entity recognition model to improve the performance of Chinese legal NER. Zhang et al. (2023) respectively used RoBERTa and Skip-Gram method to extract char-level and word-level feature of a legal document, and fused them. Dong et al. (2022) developed NERP, NERCGP, and NERFPP to expand the named entity recognition corpus in the judicial field, and introduced BERT-WWM (whole-word mask)-EXT(extra) and BiLSTM-CRF for Chinese NER task. Lee et al.

(2023) proposed NER model for Chinese legal texts, LeArNER, utilized minimal annotated data for model training to reduce expenses associated with corpus collection and training.

To sum up, NLP technology has been developing. However, due to the limitations of the text characteristics and the scarcity of judicial entity tagging corpus, the research on the processing of judgment documents by NER technology is still in its infancy (Yuan and Zhang 2021).

3 Methodology

Though various NER methods have been proposed in recent years, most of the existing methods only considered one kind of word-enhanced information, which may not be effective for the Chinese characters. In addition, there are rare NER methods applied to Chinese judgment documents, so it is necessary for us to propose a model that can be well applied in this field.

Aiming at the problem of nested entities in Chinese legal texts, we proposed a method that combines adversarial training and add word information into the NER task. It mainly includes two parts: (1) Combining character and word information to make each Chinese character provide more valuable resources for the model than a single word or character, and provide more help for NER task. (2) The adversarial training framework extractes the common features of CWS and NER task, while the noise of CWS task can be filtered. It makes the boundary of legal entities more clear and improve the effect of legal nested NER.

3.1 Model architecture

Generative Adversarial Net (GAN) (Goodfellow et al. 2020) have received considerable attention since the 2014 groundbreaking work (Wang 2020). GAN is inspired by a minimax two-player game. The optimization goal is to achieve Nash equilibrium so that the generator can estimate the distribution of data samples (Su et al. 2022). GAN is mainly used in computer vision (Mirza and Osindero 2014). In recent years, the idea of adversarial training is introduced in NLP. Different from computer vision, the input sequences in NLP are discrete values. Although the adversarial training has been successfully applied in some fields of NLP, there is little research in the judicial domain to use the idea of adversarial training to the NER task of Chinese judgment documents. In addition, the Chinese NER task and CWS task have many similar word boundaries. Most existing NER methods do not fully exploit the word boundary information of CWS.

In this paper, a novel word-enhanced model called AT-Flat is proposed that combines adversarial training and add word information into the NER task. The first dynamic framework is the adversarial training which aims to extract common features of the Chinese CWS task and NER task. The another dynamic framework is the Flat-Lattice Transformer for the NER task. With the power of Transformer and well-designed position encoding, FLAT can fully leverage the sequence information between words.

The model architecture is shown in Fig. 2. The model consists of three parts: the NER task, the CWS task and the adversarial training dynamic framework. The NER task is the goal of our methods, and the core structure is Flat. The CWS task divides sentences into words, which is similar to the NER task. The adversarial training is a dynamic framework for adding word information, including a shared BiLSTM generator, a self-attention layer and a task discriminator. In the following subsections, the parts of model will be described in detail.

3.2 Embeddings

The first step is character embedding, which aims to map discrete characters into the distribute representations. The BIO annotation method is used to label data in character form, which B represents the beginning part of the entity; I represents the middle part of the entity, and O represents the non-entity. For example, to label a person entity, the first character is labeled B-PER, and the other characters are labeled I-PER. For a given Chinese sentence $\mathbf{x} = \{c_1, c_2, \ldots, c_n\}$ from Chinese NER dataset or CWS dataset, we lookup embedding vector from pre-trained embedding matrix for each character c_i as $\mathbf{x}_i \in \mathbb{R}^{de}$. The word embedding will be introduced in the subsection Flat-Lattice Transformer.

3.3 Adversarial training

Long short term memory (LSTM) (Hochreiter and Schmidhuber 1997) is a variant of recurrent neural network (RNN) (Schuster and Paliwal 1997), which enables to address the gradient vanishing and exploding problems in RNN via introducing gate mechanism and memory cell. The unidirectional LSTM only leverages information from the past, ignoring the future information. The BiLSTM model is used in the shared generator instead of the LSTM model in AT-Flat. The reason is that BiLSTM outputs the hidden layer vectors of forward and backward, which can fully

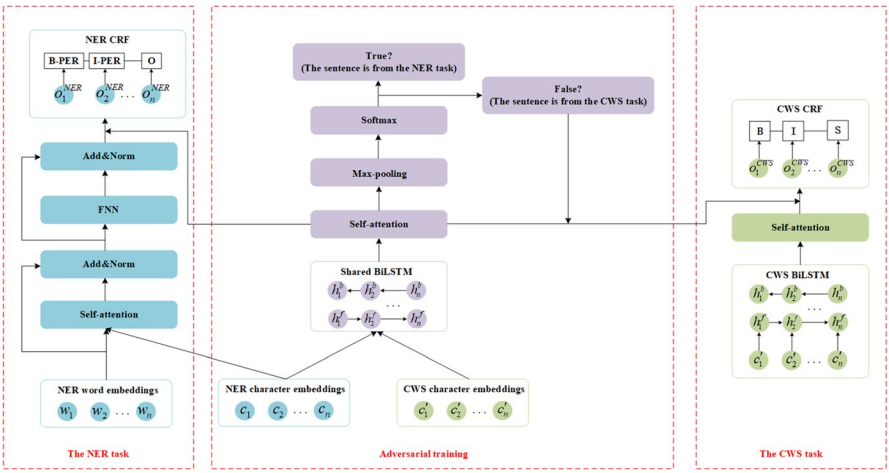

Fig. 2 The architecture of AT-Flat

integrate the contextual information. The hidden state of BiLSTM could be expressed as follows:

$$\overrightarrow{\mathrm{h}_i} = \overrightarrow{\mathrm{LSTM}}(\mathrm{x}_i, \overrightarrow{\mathrm{h}_{i-1}}), \tag{1}$$

$$\overleftarrow{\mathrm{h}_i} = \overleftarrow{\mathrm{LSTM}}(\mathrm{x}_i, \overleftarrow{\mathrm{h}_{i+1}}), \tag{2}$$

$$\mathbf{h}_i = \overrightarrow{\mathrm{h}_i} \oplus \overleftarrow{\mathrm{h}_i}, \tag{3}$$

where $\overrightarrow{\mathbf{h}_i} \in \mathbb{R}^{d_h}$ and $\overleftarrow{\mathbf{h}_i} \in \mathbb{R}^{d_h}$ are the hidden states of the forward and backward LSTM at position *i*, respectively. $\oplus$ denotes concatenation operation.

Before the output of the generator enters the discriminator, a self-attention layer is added to explicitly learn the dependencies between any two characters in sentence and capture the inner structure information of sentence (Cao et al. 2018). The functions can be expressed as

$$\mathrm{Attention}(\mathrm{Q}, \mathrm{K}, \mathrm{V}) = \mathrm{softmax}(\frac{\mathrm{QK}^T}{\sqrt{\mathrm{d}}})\mathrm{V}, \tag{4}$$

$$head_i = \mathrm{Attention}(\mathrm{QW}_i^Q, \mathrm{KW}_i^K, \mathrm{VW}_i^V), \tag{5}$$

$$\mathrm{H}' = (head_i \oplus \ldots head_h)\mathrm{W}_o, \tag{6}$$

where $\mathbf{Q} \in \mathbb{R}^{N \times 2d_h}$ is query matrix, $\mathbf{K} \in \mathbb{R}^{N \times 2d_h}$ is key matrix and $\mathbf{V} \in \mathbb{R}^{N \times 2d_h}$ is value matrix. The four matrix are set $\mathbf{Q} = \mathbf{K} = \mathbf{V} = \mathbf{H}$. $\mathbf{d}$ is the dimension of hidden units of BiLSTM, and it is usually set to be $2d_h$. $\mathbf{W}_i^Q \in \mathbb{R}^{2d_h \times d_k}$, $\mathbf{W}_i^K \in \mathbb{R}^{2d_h \times d_k}$, $\mathbf{W}_i^V \in \mathbb{R}^{2d_h \times d_k}$ and $\mathbf{W}_o \in \mathbb{R}^{2d_h \times d_k}$. The four parameters are trainable. d_k is usually set to be $2d_h/h$.

The discriminator includes a max-pooling layer and a softmax layer:

$$\mathbf{s}'^k = \mathrm{Maxpooling}(\mathbf{S}'^k), \tag{7}$$

$$D(\mathbf{s}'^k; \theta_d) = \mathrm{softmax}(\mathbf{W}_d \mathbf{s}'^k + \mathbf{b}_d), \tag{8}$$

where $\mathbf{s}'^k$ is the output of the self-attention layer after the shared BiLSTM generator of task *k*. θ_d is the parameter of the discriminator. $\mathbf{W}_d \in \mathbb{R}^{K \times 2d_h}$ and $\mathbf{b}_d \in \mathbb{R}^K$. These two parameters are also trainable. *K* is the number of tasks. $D(\mathbf{s}'^k; \theta_d)$ is the output of the discriminator, which can be seen as a softmax function. The softmax function is a standard mapping method that converts values into probabilities. The activation function uses softmax to make the discriminator output the probability that the sentence comes from the NER task.

The adversarial loss L_{adv} is introduced to prevent specific features of CWS task from creeping into shared space. The optimization function can be expressed as

$$L_{adv} = \min_{\theta_s}(\max_{\theta_d} \sum_{k=1}^{K} \sum_{i=1}^{T_k} \log D(\mathbf{s}'^k; \theta_d)), \tag{9}$$

where θ_s is the trainable parameter of the shared BiLSTM generator. T_k is the number of training sentences. Therefore, the adversarial training becomes a minimax optimization problem. The adversarial loss encourages the shared encoder to learn task-invariant features such that the task discriminator cannot reliably determine whether a sample originates from the NER or CWS task. These shared features capture information that is crucial for identifying entity boundaries in the NER task. Importantly, the goal is not to enforce alignment with CWS segmentation, but to promote representations that remain robust under perturbation and are semantically consistent across tasks.For example, as shown in Fig. 3, given a sentence "被告承担主要责任(defendant assumes primary responsibility)", the NER task and the CWS task have the same boundary for some words such as "被告(defendant)", "承担(assumes)", while the boundary in NER task is "主要责任(primary responsibility)" and the boundary of CWS task is "主要(primary)", " 责任(responsibility)". In such cases, the adversarial framework does not force the model to follow the CWS segmentation. Instead, it encourages the model to retain the span "主要责任(primary responsibility)" as a unified representation if doing so leads to more stable and consistent features, thereby favoring boundary patterns that benefit NER performance.

In the whole processing of the adversarial training, the character embedding of the NER task and CWS task are alternately input into the shared BiLSTM generator. Then, the multi-head self-attention mechanism is used to explicitly learn the dependencies between any two characters in a sentence and capture the inner structure information of a sentence. The output will be sent into the discriminator. If the discriminator determines that the input is from the NER task, the output of the self-attentive layer is sent to a independent NER task for subsequent processing. If the input is determined to be from the CWS task, the output is sent to a independent CWS task structure. There is no output labeling for the CWS task in this paper because we focuse on the Chinese NER task. The character embeddings of the two tasks alternate input and continue to conduct the adversarial training. Finally, the shared BiLSTM generator will generate the result that the discriminator cannot distinguish which task the input comes from. The result is the common features learned by the shared BiL-

Fig. 3 An example of describing the similarities and differences between Chinese NER and CWS

STM. The common features contain the task-shared word boundaries, which is useful for the Chinese NER task to recognize the boundary of the entities.

3.4 Flat-lattice transformer

Focusing on the NER task, the flat-lattice architecture (Li et al. 2020) uses Transformer encoder. It is mainly composed of self-attention and feedforward network (FFN) layers. Each sublayer is followed by residual connection and layer normalization. FFN is a position-wise multi-layer perceptron with non-linear transformation.

The flat-lattice is defined as a set of spans, and a span corresponds to a token, a head and a tail, like in Fig. 4. The token is a character or word. The head and tail represent the position index of the token's first and last characters in a sequence, and they indicate the position of the token in the lattice. The head and tail are the same for each character.

The flat-lattice structure consists of spans with different lengths. The relative position encoding of spans is used to encode the interactions among spans. For two spans x_i and x_j in the lattice, according to their heads and tails, there are three kinds of relations between them: intersection, inclusion and separation. A dense vector is used to represent the relationships, and it is calculated by continuous transformations of the head and tail information. So it can not only represent the relation between two tokens, but also show more detailed information, such as the distance between words. Four kinds of relative distances can be calculated with x_i and x_j:

$$d_{ij}^{(hh)} = head[i] - head[j], \tag{10}$$

$$d_{ij}^{(ht)} = head[i] - tail[j], \tag{11}$$

$$d_{ij}^{(th)} = tail[i] - head[j], \tag{12}$$

$$d_{ij}^{(tt)} = tail[i] - tail[j], \tag{13}$$

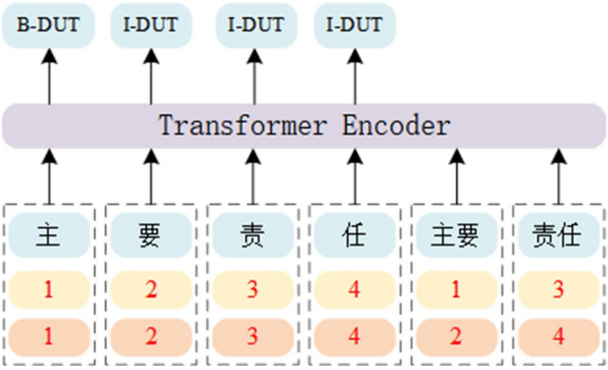

Fig. 4 Flat-Lattice Transformer

where *head*[i] and *tail*[i] are the head and tail position of span x_i; $d_{ij}^{(hh)}$ is the distance between head of x_i and tail of x_j, and the meanings of $d_{ij}^{(ht)}$, $d_{ij}^{(th)}$ and $d_{ij}^{(tt)}$ are similar. The final relative position encoding of spans is a non-linear transformation of the four distances:

$$R_{ij} = \mathrm{ReLU}(W_r(\mathbf{p}_{d_{ij}^{(hh)}} \oplus \mathbf{p}_{d_{ij}^{(th)}} \oplus \mathbf{p}_{d_{ij}^{(ht)}} \oplus \mathbf{p}_{d_{ij}^{(tt)}})), \tag{14}$$

where W_r is a learnable parameter, $\oplus$ denotes the concatenation operator, and $\mathbf{p}_d$ is calculated as in Vaswani et al. (2017):

$$\mathbf{p}_d^{(2k)} = \sin(d/10000^{2k/d_{model}}), \tag{15}$$

$$\mathbf{p}_d^{(2k+1)} = \cos(d/10000^{2k/d_{model}}), \tag{16}$$

where d is $d_{ij}^{(hh)}$, $d_{ij}^{(ht)}$, $d_{ij}^{(th)}$ or $d_{ij}^{(tt)}$ and k is the index of dimension of position encoding. Then a variant of self-attention (Dai et al. 2019) is used to leverage the relative span position encoding:

$$\mathbf{A}_{i,j}^* = \mathbf{W}_q^\top \mathbf{E}_{x_i}^\top \mathbf{E}_{x_j} \mathbf{W}_{k,E} + \mathbf{W}_q^\top \mathbf{E}_{x_i}^\top \mathbf{R}_{ij} \mathbf{W}_{k,R} + \mathbf{u}^\top \mathbf{E}_{x_j} \mathbf{W}_{k,E} + \mathbf{v}^\top \mathbf{R}_{ij} \mathbf{W}_{k,R}, \tag{17}$$

where $\mathbf{W}_q$, $\mathbf{W}_{k,R}$, $\mathbf{W}_{k,E} \in \mathbb{R}^{d_{model} \times d_{head}}$ and u, $\mathbf{v} \in \mathbb{R}^{d_{head}}$ are learnable parameters. The following calculation is the same to the vanilla Transformer. The architecture of FLAT is shown in Fig. 5.

3.5 Conditional random field of the NER task

Considering the dependence between tags, Conditional random field (CRF) (Lafferty et al. 2001) is used to output the final entity prediction results. The CRF layer can add some constraints to the predicted labels to ensure that they are legal. For example, the first word in a sentence should always start at the label "B-" or "O", not "I-". Due to the difference in the tasks, a specific CRF layer is introduced for each task. Due to the difference in the tasks, a specific CRF layer is used for each task.

For a sentence k, the final representation via concatenating the representations from NER and shared module after self-attention layer is computed:

$$\mathbf{H}''^k = \mathbf{H}'^k \oplus \mathbf{S}'^k, \tag{18}$$

where $\mathbf{H}'^k$ and $\mathbf{S}'^k$ are the outputs of NER module and shared module of task k, respectively.

Given a sentence $\mathbf{x} = \{c_1, c_2, \ldots, c_n\}$ and a sequence $\mathbf{y} = \{y_1, y_2, \ldots, y_n\}$ with predicted tags, the CRF tagging process can be described as follows:

$$\mathrm{o}_i = \mathrm{W}_s \mathrm{h}_i'' + \mathrm{b}_s, \tag{19}$$

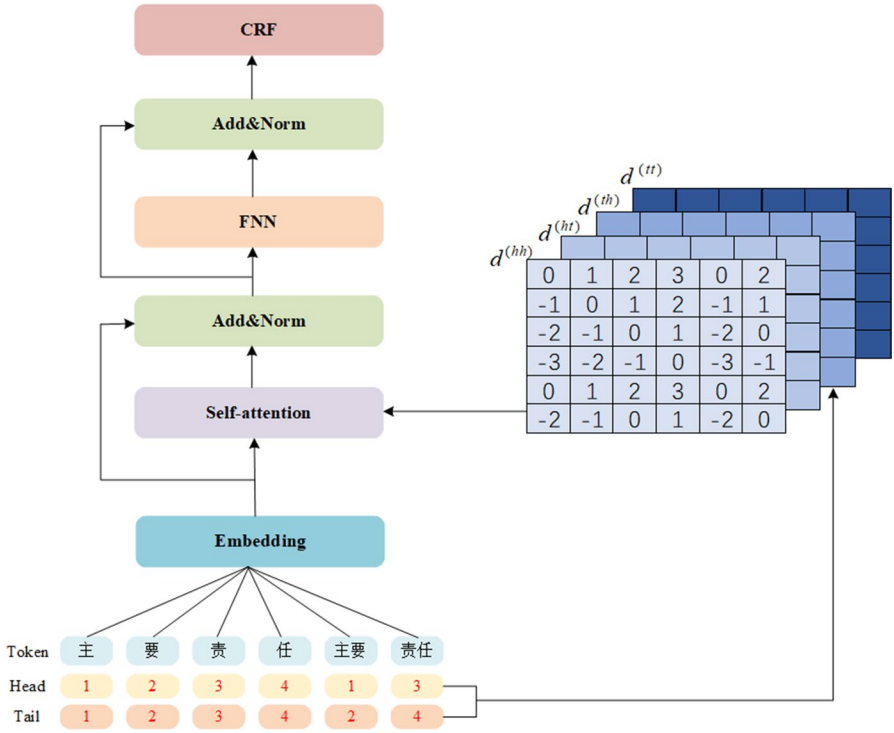

Fig. 5 The architecture of FLAT

$$s(\mathrm{x}, \mathrm{y}) = \sum_{i=1}^{N} (\mathrm{o}_{i,y_i} + \mathrm{T}_{y_i-1,y_i}), \tag{20}$$

$$\bar{\mathbf{y}} = \arg\max_{\mathbf{y} \in \mathbf{Y}_x} s(\mathrm{x}, \mathrm{y}), \tag{21}$$

where $\mathbf{W}_s \in \mathbb{R}^{|T| \times 4d_h}$ and $\mathbf{b}_s \in \mathbb{R}^{|T|}$ are trainable parameters. $|T|$ is the number of output labels. $\mathbf{o}_{i,y_i}$ represents the score of the y_i-th tag of the character c_i. T is a transition score matrix which defines the scores of two successive labels. $\mathbf{Y}_x$ is all candidate tag sequences for given sentence x. In decoding, the Viterbi algorithm is used to get the predicted tag sequence $\bar{\mathbf{y}}$. The probability of the true label sequence can be calculated as follows:

$$p(\hat{\mathrm{y}}|\mathbf{x}) = \frac{e^{s(\mathrm{x}, \hat{\mathrm{y}})}}{\sum_{\tilde{\mathrm{y}} \in \mathrm{Y}_x} e^{s(\mathrm{x}, \tilde{\mathrm{y}})}}, \tag{22}$$

where $\hat{\mathrm{y}}$ is the ground-truth label sequence. Given T training examples $(\mathrm{x}^{(i)}; \hat{\mathrm{y}}^{(i)})$, the loss function L_{Task} can be defined:

$$L_{Task} = -\sum_{i=1}^{T} \log p(\hat{\mathrm{y}}^{(i)}|\mathrm{x}^{(i)}), \tag{23}$$

The back gradient propagation method is used to minimize the loss function. According to the adversarial training, the loss function L_{adv} is computed. According to the independent NER and CWS task, the loss function L_{Task} is computed. So the final loss function of the model can be expressed:

$$\begin{aligned} L &= L_{NER} \cdot I(\mathbf{x}) + L_{CWS} \cdot (1 - I(\mathbf{x})) + \lambda L_{adv}, \\ I(\mathbf{x}) &= \begin{cases} 1, & if \quad \mathbf{x} \in D_{NER}; \\ 0, & if \quad \mathbf{x} \in D_{CWS}, \end{cases} \end{aligned} \tag{24}$$

where λ is a hyper-parameter. L_{NER} and L_{CWS} can be obtained via (23). $I(\mathbf{x})$ is a switching function to identify which task the input comes from. In training for each iteration, the NER and CWS tasks are selected in turn. Then a batch of training examples is selected from a given task to update the parameters. Since Chinese NER task and CWS task may have different convergence rate, the total iteration number can be chosen according to the performance of the NER task.

4 Experiments

In this section, experiments of the proposed model are showed with real Chinese judgment documents and public Chinese NER datasets. It includes five subsections: datasets, parameter setting, evaluation metrics, baselines and experimental results.

4.1 Datasets

In this paper, we conducted our experiments on 500 traffic accident judgment documents, and 3,300 paragraphs from financial lending judgment documents. As shown in Table 1, 20 types of named entity in traffic accident judgment documents is defined. Financial lending judgment documents have 12 types named entities in Table 2.

In addition, three public Chinese nested NER datasets[1] were used, including (1) Bank, a bank lending dataset; (2) ECommerce, a named entity recognition dataset for e-commerce domain (3) Fned, a dataset containing different domain event detection. Figure 6 shows the examples of nested entities in these public datasets. In the experiments, 70% of the datasets is randomly chosen as the training set and the other is taken as the test set. Some experiments further adopt 10-fold cross-validation on the training set to ensure robust evaluation.

4.2 Parameter setting

Hyperparameters are summarized in Table 3, and most of them are common setting in many references (e.g., Li et al. 2020).

[1] https://github.com/liucongg/NLPDataSet

Table 1 Statistic of traffic accident dataset

Label	Entity type	Details	Amount of entity
NNP	非自然人主体	Non-natural person	427
NCF	行人未走人行横道或过街设施	The pedestrian is not using a crosswalk or street crossing	8
INS	保险类别	Insurance	1623
DIS	伤残	Disability	524
MV	机动车	Motor vehicle	2064
NMV	非机动车	Non-motor vehicle	192
VTL	违反道路交通信号灯	Violation of road traffic lights	4
ICL	违法变更车道	Illegal lane change	3
DEA	死亡	Death	247
UD	未取得驾驶资格	Not obtaining driving qualifications	40
GI	一般人身损害	General personal injury	830
DUT	责任认定	Duty confirmation	1097
OVE	超载	Overload	14
NP	自然人主体	Natural person	11929
PLC	财产损失赔偿项目	Property loss compensation project	382
DAD	饮酒后驾驶	Drivingafterdrinking	17
DD	醉酒驾驶	Drunkendriving	31
PIC	人身损害赔偿项目	Personal injury compensation project	3418
SPE	超速	Overspeed	11
NAP	不避让行人	Not avoid pedestrians	6

4.3 Evaluation metrics

Three widely-adopted metrics are used to evaluate our method, i.e., Precision (P), Recall (R) and F_1 score:

$$\text{Precision} = \frac{1}{|C|}\sum_{i \in C}\frac{(TP_i + TN_i) \times TP_i}{TP_i + FP_i}, \tag{25}$$

$$\text{Recall} = \frac{1}{|C|}\sum_{i \in C}\frac{(TP_i + TN_i) \times TP_i}{TP_i + FN_i}, \tag{26}$$

$$F_1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}, \tag{27}$$

where TP_i is the number of instances correctly assigned to one class; FP_i is the number of instances which belong to other classes but are assigned to that class; FN_i is the number of instances that belong to that class but are mistakenly assigned to other classes. C represents the number of all categories. F_1 value is the harmonic mean of the Precision and Recall.

4.4 Baseline

We compare AT-Flat with five state-of-the-art methods for the NER task, including BiLSTM-CRF (Huang et al. 2015), which is the most widely used basic model for the NER task; BiLSTM-Softmax; Att-BiLSTM (Zhou et al. 2016), which is a BiLSTM-CRF enhanced model; BiLSTM + CRF + adversarial + self-attention (Cao et al. 2018), and Flat-Lattice Transformer (Li et al. 2020).

4.5 Results and analysis

4.5.1 Traffic accident dataset

We compare our model AT-Flat with other baselines, and the results in traffic accident dataset can be seen from Table 4, where the best method is marked in bold. Overall, AT-Flat achieves the best performance on the recall rate of 96.07%, and F_1 score of 95.35%. Compared to the worst-performing BiLSTM-Softmax, AT-Flat improves at least 3% in each metric. These metrics of the proposed model also significantly increased compared to BiLSTM-CRF. Att-BiLSTM contains the attention mechanism but without introducing adversarial training and word information, which is probably the reason that its performance is worse than AT-Flat. Similarly, Flat does not include adversarial training, and the model with adversarial training (BiLSTM + CRF + adversarial + self-attention) do not contain word information, which makes their performance weaker. The reason why these methods are not effective may be that they do not identify the boundaries of legal entities well and divide many entities from the inside. These results show that AT-Flat combines the information of characters and words, and expands the range of recognized legal entities, so it can obtain

Table 2 Statistic of financial lending dataset

Label	Entity type	Details	Amount of entity
PER	人名	Person's name	1522
CB	共同借款人	Co-borrower	61
CBLE	提前收回借款	Collect back of loans early	17
CP	共同还款人	Co-payor	49
JLG	连带责任保证	Joint liability guarantee	62
LA	贷款数额	Loan amount	164
LO	贷款	Loan	279
LP	贷款用途	Loan purpose	39
MG	最高额保证	Maximum guarantee	11
MM	最高额抵押	Maximum mortgage	8
PA	抵押物	Pawn	73
TC	解除合同	Contract termination	1

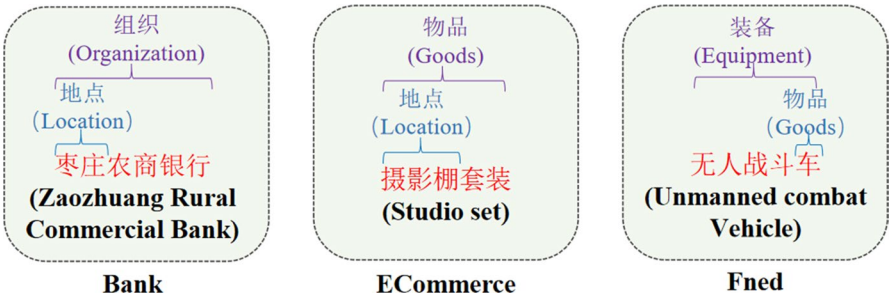

Fig. 6 Nested entities in public datasets

Table 3 Hyperparameter values

Description	Value
Size of character embedding	50
Size of Flat word embedding	50
Batch size	10
Size of the LSTM	105
Embed dropout	0.5
Learning rate	8e-04
Warmup	0.01
Optimizer	SGD
Head	8

more information of Chinese judgment documents and get better recognition results. In addition, the adversarial training framework can learn task-shared word boundary information from CWS task.

To make the results more reliable, we also conducted the experiments of 10-fold cross-validationand, and the results are shown in Table 5. Although BiLSTM-CRF

Table 4 Comparison of experimental results on traffic accident dataset

Models	P%	R%	F1%
BiLSTM-CRF	93.95	93.63	93.79
BiLSTM-Softmax	91.46	93.00	92.22
Att-BiLSTM	92.28	93.22	92.75
BiLSTM+CRF+adversarial+self-attention	94.49	94.92	92.80
Flat	94.19	94.78	93.02
AT-Flat	**94.64**	**96.07**	**95.35**

Table 5 10-fold cross-validation experimental results on traffic accident dataset

Models	P%	R%	F1%
BiLSTM-CRF	**94.69**	91.46	93.05
BiLSTM-Softmax	91.79	93.39	92.58
Att-BiLSTM	91.67	93.40	92.52
BiLSTM+CRF+adversarial+self-attention	93.79	94.47	92.43
Flat	94.88	94.91	93.05
AT-Flat	94.58	**95.83**	**95.20**

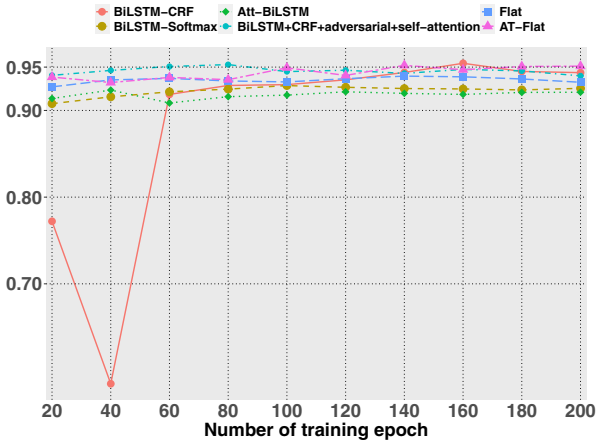

Fig. 7 The Precision of each method with different epochs on traffic accident dataset

got the highest Precision of 94.69%, AT-FLAT is comparable to it with the values of 94.58%. Meanwhile, AT-FLAT achieved the highest Recall of 95.83% and the highest F_1 of 95.20%, respectively. It was also much better than the second one. In general, compared with other methods, AT-Flat performs well in all three metrics, which confirms the stronger stability of our method.

In additon, Figs. 7, 8, and 9 show the metrics of each method with different epochs. It can be seen that our model is more stable and has better performance in the iterative process, especially on Recall (Fig. 8). Among these methods, the training speed of BiLSTM-CRF is slowest, and the results are worst. Although Att-BiLSTM introduces the attention mechanism, the results are not so different from BiLSTM-Softmax. It shows that only adding the attention layer does not significantly improve the results of NER. Flat adds word information, so it works better compared to other BiLSTM methods, which indicating that word information is important in the NER task. BiLSTM + CRF + adversarial + self-attention is second only to AT-Flat, which

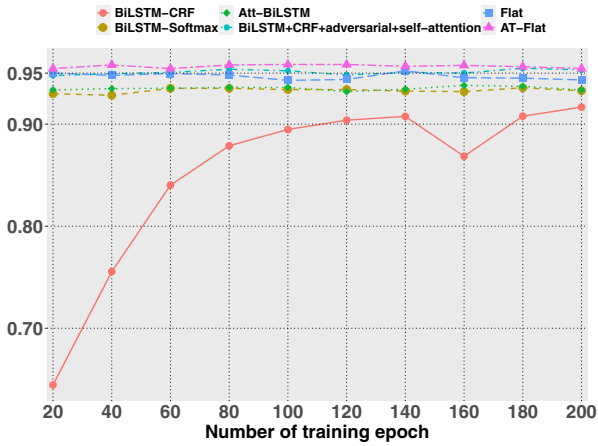

Fig. 8 The Recall of each method with different epochs on traffic accident dataset

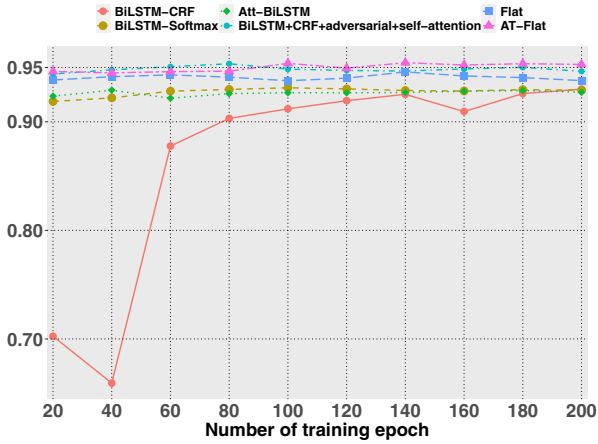

Fig. 9 The F_1 of each method with different epochs on traffic accident dataset

indicating that adversarial training with CWS and NER can improve the accuracy of NER. However, the training process without word information, it is not as effective as AT-Flat. In summary, AT-Flat adds adversarial training and word information can effectively improve the accuracy of the NER task, and achieve more than 95% in all metrics within 200 iterations.

4.5.2 Financial lending dataset

Table 6, and 7 and Figs. 10, 11, and 12 show the results of each method on financial lending dataset. It can be seen that AT-Flat still performs well on Recall and F_1, and they were stabilized after training 80 epochs. BiLSTM-CRF, Att-BiLSTM and BiLSTM-Softmax are significantly less effective on this dataset. Although AT-Flat's performance on Precision is not the best, it has a tendency to grow slowly as the number of training epochs increases (Fig. 10). The Recall of AT-Flat is markedly superior to other methods (Fig. 11), so its F_1 value is no less impressive (Fig. 12).

Table 6 Comparison of experimental results on financial lending dataset

Models	P%	R%	F1%
BiLSTM-CRF	58.30	52.49	55.24
BiLSTM-Softmax	39.31	42.18	40.69
Att-BiLSTM	40.61	46.24	43.25
BiLSTM+CRF+adversarial+self-attention	62.51	66.69	64.53
Flat	**63.09**	67.17	65.07
AT-Flat	59.68	**73.95**	**66.05**

Table 7 10-fold cross-validation experimental results on financial lending dataset

Models	P%	R%	F1%
BiLSTM-CRF	64.23	65.01	64.60
BiLSTM-Softmax	50.68	59.71	54.81
Att-BiLSTM	49.34	60.15	54.13
BiLSTM+CRF+adversarial+self-attention	**65.61**	62.44	63.91
Flat	63.69	68.85	66.17
AT-Flat	62.59	**73.92**	**67.72**

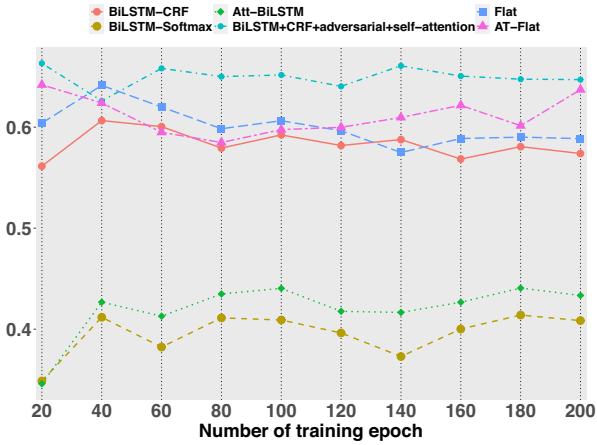

Fig. 10 The Precision of each method with different epochs on financial lending dataset

Surprisingly, Att-BiLSTM and BiLSTM-Softmax got less than 50% of the metrics When 200 epochs were trained, which is completely different from the results of the traffic accident dataset. Methods other than ours also show less stability in this dataset. Financial lending and traffic accident are different fields of Chinese judgment documents. Therefore, the features and the types of entity are various, which affects the experimental results. There are also some special terms in the legal field that may make it difficult for normal models to accurately identify entity boundaries.

We observe a substantial performance drop on this dataset compared to other domains, which we attribute primarily to limited training data. Although AT-Flat improves the recognition of nested entities compared to baselines, some challenges persist, including:

•Lengthy and compound entities: For example, the model failed to correctly identify the full span of entities like "中国建设银行股份有限公司深圳分行龙岗支行" ("China Construction Bank Shenzhen Branch Longgang Sub-branch"), predicting

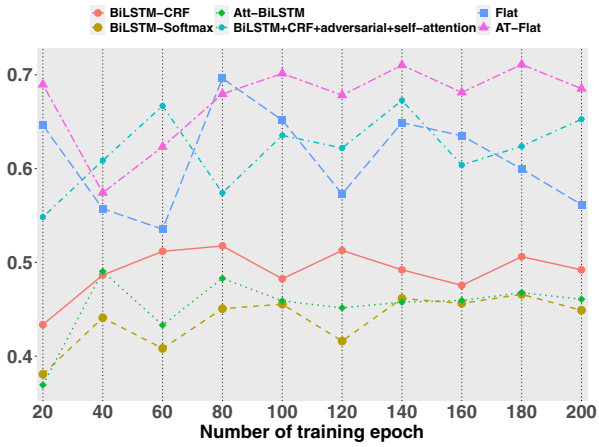

Fig. 11 The Recall of each method with different epochs on financial lending dataset

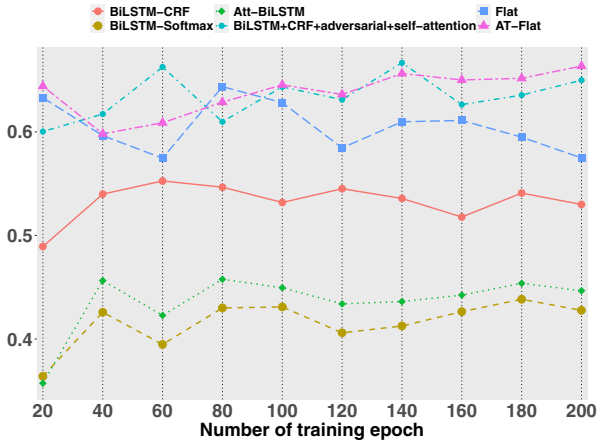

Fig. 12 The F_1 of each method with different epochs on financial lending dataset

only a partial entity. Such long and hierarchical organization names challenge the boundary modeling capacity of AT-Flat.

•Domain-specific terminology: Specialized legal and financial terms such as "保证人" (guarantor) and "履约责任" (performance obligation) are occasionally misclassified or missed due to subtle contextual cues and vocabulary sparsity.

These observations indicate that the performance degradation arises not merely from dataset size but also from the inherent complexity of the domain. Future work will explore integrating domain adaptation techniques and structure-aware models to better address these challenges.

4.5.3 Public datasets

The experimental results of three public Chinese NER datasets are shown in Table 8, and 9. Overall, our proposed model still has steady and superior results. A interesting conclusion can be observed that blindly increasing the complexity of the model does not necessarily get good results with comparing the results. Att-BiLSTM intro-

Table 8 Comparison of experimental results on public datasets

	Bank			ECommerce			Fned		
Models	P%	R%	F1%	P%	R%	F1%	P%	R%	F1%
BiLSTM-CRF	82.53	81.77	82.14	81.98	81.46	81.72	75.15	72.95	74.03
BiLSTM-Softmax	76.07	79.51	77.75	73.12	79.19	76.03	67.29	70.75	68.97
Att-BiLSTM	72.01	74.66	73.30	70.95	76.49	73.61	62.74	66.64	64.62
BiLSTM+CRF+adversarial +self-attention	83.23	82.47	82.84	81.59	84.19	82.85	**77.47**	78.26	77.85
Flat	83.24	81.44	82.33	81.74	83.86	82.74	76.22	83.15	79.51
AT-Flat	**83.25**	**85.89**	**84.55**	**82.58**	**83.96**	**83.25**	77.01	**83.74**	**80.22**

Table 9 10-fold cross-validation experimental results on public datasets

	Bank			Ecommerce			Fned		
Models	P%	R%	F1%	P%	R%	F1%	P%	R%	F1%
BiLSTM-CRF	83.09	81.56	82.31	84.07	83.49	83.77	75.27	73.39	74.31
BiLSTM-Softmax	76.50	79.60	78.01	76.01	81.64	78.72	67.33	71.09	69.16
Att-BiLSTM	72.92	75.48	74.17	73.66	78.67	76.08	62.89	66.83	64.79
BiLSTM+CRF+adversarial +self-attention	83.63	81.56	82.58	82.58	83.96	83.25	**77.82**	78.72	78.26
Flat	83.41	83.35	83.37	83.09	84.71	83.85	75.79	82.54	79.01
AT-Flat	**85.02**	**85.18**	**84.99**	**84.34**	**85.54**	**84.93**	76.44	**84.07**	**80.06**

duces the attention module, but the performance was worse than BiLSTM-CRF and BiLSTM-Softmax. The difference is AT-Flat adds adversarial network and introduces word-character information can improve the effectiveness of named entity recognition.

In general, the performance of AT-Flat on these datasets in different fields can verify: (1) The increase of word and character information does contribute to the enhancement of the dataset, thus affecting the entity recognition effect. (2) Adversarial training helps to determine the entity boundaries, thus making AT-Flat better applicable to different domains.

5 Conclusion

In this paper, we propose AT-Flat, an adversarial training model integrates task-shared word boundary information from CWS to improve Chinese judgment documents NER performance, and combines character and word information to get more legal entities information. Extensive experiments based on two different Chinese judgment document datasets and three public Chinese NER datasets show that AT-Flat achieves the best performance compared to the state-of-the-art models. For future work, We will adjust the model to expand its application fields.

Although AT-Flat achieves strong performance, it still has some limitations. One potential bottleneck is its limited capacity to model long-range dependencies, which may affect its effectiveness when processing lengthy legal documents. Additionally, the model's inference efficiency might degrade as input length increases, posing chal-

lenges for real-world deployment in legal or industrial scenarios. We leave addressing these issues for future work.

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Author Contributions Jiabao Wang and Yang Weng conceived the study. Jiabao Wang designed the algorithm, and Kaixuan Wang performed the computation. The analysis and manuscript was written by Jiabao Wang. Jiabao Wang, Yang Weng, and Xin Li revised the manuscript.

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